

Exact Kernel Transfer to Clique Complexes and the Hardness of Normalized Persistence

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Abstract

For an inclusion of clique complexes $X_1 \subseteq X_2$, the normalized persistence in degree d is the fraction

$$\frac{\text{rank}[H_d(X_1) \rightarrow H_d(X_2)]}{\dim H_d(X_1)}$$

of the degree- d holes of X_1 that survive in X_2 . Estimating this fraction requires controlling the exact homology at both endpoints and the map induced by inclusion, even when both endpoint Laplacians have inverse-polynomial spectral gaps. We prove that estimating the fraction to additive error $1/24$ is hard for $\text{BQP}_1^{G_2}$, the perfect-completeness class over the exact gate set $G_2 = \{X, CX, CCX, H \otimes H\}$, and hence for BQP_1 over every finite gate set with entries in a cyclotomic field $\mathbb{Q}(\zeta_{2^k})$, even for unweighted clique complexes. The main tool is a finite-certificate kernel-transfer theorem. For a fixed palette of weighted clique gadgets satisfying finitely many exactly checkable local conditions, every unit chain x of the full geometric complex satisfies

$$\text{dist}(x, K)^2 \leq C \left(t\lambda^2 + \frac{\langle x, \Delta x \rangle}{g\lambda^{26}} \right),$$

where K is the embedded kernel of the simulated projector Hamiltonian, g its gap, t the number of gadgets, and λ the private vertex weight. Since λ is chosen independently of g , the geometric Laplacian has exactly $\dim K$ zero modes and a gap linear in g above them. Exact fillings identify the endpoint homology with a quotient V/W_A of the register cycle space, and nested term sets induce the natural quotient epimorphisms, so the persistent rank equals the later kernel dimension without any choice of compatible harmonic representatives. A fixed eight-dimensional label register turns a $\text{BQP}_1^{G_2}$ verifier into instances with $\beta_d(X_1) = 8$ and normalized persistence $3/4$ or $1/8$, and an established common-copy blow-up transfers everything to unweighted graphs.

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1 Introduction

Persistent homology records which topological features of a growing simplicial complex survive as the complex grows [7, 24, 4]. For a two-step filtration $X_1 \subseteq X_2$ and a degree d , the basic invariant is the persistent Betti number, the rank of the induced map $H_d(X_1) \rightarrow H_d(X_2)$. Dividing it by $\beta_d(X_1)$ gives a scale-free and directly interpretable

37 statistic,

$$38 \quad q_d(X_1, X_2) = \frac{\text{rank}[H_d(X_1) \rightarrow H_d(X_2)]}{\beta_d(X_1)} \in [0, 1], \quad (1)$$

39 the fraction of the degree- d holes of X_1 that are still present in X_2 . Recent work [19]
 40 introduced this quantity under the name *normalized harmonic persistence* and asked
 41 for its complexity on clique complexes of weighted graphs, the input model of quantum
 42 topological data analysis [18, 10, 8, 2, 20]. The motivation is that the known quantum
 43 algorithms for Betti and persistent Betti numbers return quantities normalized by the
 44 number of simplices [18, 10, 23, 1], whereas a fraction of holes is the statistic one would
 45 actually ask for. Under inverse-polynomial spectral-gap promises on the two endpoint
 46 Laplacians, that work proved that a low-energy relaxation of q_d is DQC_1 -hard and
 47 in BQP , conjectured that q_d itself is hard for a perfect-completeness variant SDQC_1
 48 of DQC_1 (Conjecture 1 of [19]), and reduced this conjecture to a *coefficient-sensitive*
 49 *kernel realization* of history Hamiltonians by clique complexes (Conjecture 2 of [19]).
 50 That work identifies a kernel-preserving simulation of frustration-free Hamiltonians by
 51 combinatorial Laplacians as the missing ingredient and an important open question.

52 Why exact kernels are hard to realize.

53 The exact-kernel primitive available in the literature is a clique-gadget construction [5,
 54 15]: a graph whose clique complex has one hole per computational basis state, together
 55 with one weighted gadget per local projector of a frustration-free Hamiltonian H . The
 56 gadgets produce the desired zero modes, and the many-gadget spectral analysis [15]
 57 proves a gap when the simulated H is positive definite. A persistence reduction needs
 58 more. It needs two Hamiltonians H_{in} and $H_{\text{out}} = H_{\text{in}} + H_{\text{rej}}$ with known but generally
 59 large kernels, the exact multiplicity $\dim \ker \Delta = \dim \ker H$ at both endpoints, an
 60 inverse-polynomial gap above the *complete* kernel at both endpoints, and control of
 61 the inclusion-induced map. One must rule out every additional geometric zero mode
 62 and every small positive eigenvalue, including for chains supported almost entirely on
 63 private gadget vertices, with constants that do not depend on the exponentially large
 64 padded register.

65 Results.

66 We carry out this realization for the history Hamiltonians of exact circuits over the gate
 67 set $G_2 = \{X, CX, CCX, H \otimes H\}$, and obtain, to our knowledge, the first unconditional
 68 hardness theorem for the exact fraction (1) on clique complexes. Under the exact-
 69 circuit convention of [22], $\text{BQP}_1^{G_2}$ is the class of promise problems decided with perfect
 70 completeness by uniform exact G_2 circuits; by the exact simulation theorem in [22] it
 71 contains BQP_1^G for every finite gate set G with entries in a cyclotomic field $\mathbb{Q}(\zeta_{2^k})$, for
 72 instance Clifford+ T .

73 ► **Theorem 1.1** (Main theorem; Theorem 5.3 and Corollary 6.1). *There is a deterministic*
 74 *polynomial-time reduction that maps an instance x of any $\text{BQP}_1^{G_2}$ promise problem L*
 75 *to nested unweighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ on a common vertex set, a degree d ,*

and rationals $\gamma_{\text{in}}, \gamma_{\text{out}} \geq 1/\text{poly}(|x|)$ such that $\Delta_{X_{\text{in}},d}$ and $\Delta_{X_{\text{out}},d}$ have no eigenvalue in $(0, \gamma_{\text{in}})$ and $(0, \gamma_{\text{out}})$ respectively, $\beta_d(X_{\text{in}}) = 8$, and

$$q_d(X_{\text{in}}, X_{\text{out}}) = \begin{cases} 3/4, & x \in L, \\ 1/8, & x \notin L. \end{cases}$$

Consequently, estimating q_d to additive error $1/24$ with success probability $2/3$, for weighted or unweighted clique complexes with inverse-polynomial endpoint gaps promised (Problem 9 of [19]), is $\text{BQP}_1^{G^2}$ -hard.

The reduction has three layers, each of independent interest.

► **Theorem 1.2** (Kernel transfer; Theorem 3.3, informal). *Fix a finite palette of weighted clique gadgets satisfying finitely many exactly checkable local conditions (Assumption 3.1). Attach t gadgets with private vertex weight λ to the register, simulating $H = \sum_j \Pi_j$ with $H \succeq g(P_V - P_K)$, $K = \ker H$. Then every unit chain x of the full clique complex satisfies*

$$\text{dist}(x, K)^2 \leq C \left[t\lambda^2 + \frac{\langle x, \Delta x \rangle}{g\lambda^\kappa} \right], \quad \kappa = 4m + 2, \quad (2)$$

where m bounds the number of active qubits of a gadget and C depends only on the palette. Hence, for $\lambda \leq c \min\{t^{-1}, \eta/\sqrt{t}\}$ and $E = c'\eta^2 g\lambda^\kappa$,

$$\dim \ker \Delta = \dim K, \quad \text{spec}(\Delta) \cap (0, E) = \emptyset,$$

including the case $K = 0$.

Equation (2) concerns arbitrary geometric chains, not only vectors of the simulated register, and its constants depend only on the palette. Its shape matters. The structural error $t\lambda^2$ does not involve g , so the gadget scale λ is chosen from the number of gadgets alone, independently of the logical gap, and the resulting geometric gap E is linear in g . A direct use of a many-gadget inequality in which λ is proportional to g would instead produce a high power of g . For the palette used here $m = 6$, and a conservative choice gives $E = \Omega(\eta^{28} g/t^{26})$; the exponent is poor, but the bound is inverse polynomial.

► **Theorem 1.3** (Quotient naturality; Theorem 4.2). *Under the hypotheses of Theorem 1.2, $H_d(X_A) \cong V/W_A$ with $W_A = \sum_{j \in A} \text{ran } \Pi_j = K_A^\perp$, and for $A \subseteq B$ the inclusion $X_A \subseteq X_B$ induces the natural quotient map $V/W_A \twoheadrightarrow V/W_B$. In particular $\text{rank}[H_d(X_A) \rightarrow H_d(X_B)] = \dim K_B$.*

Theorems 1.2 and 1.3 give exactly the two properties that the kernel-realization conjecture [19, Conjecture 2] requests, kernel fidelity with computable gaps and functoriality along the filtration, for the exact history Hamiltonians treated here; Section 7 states the precise comparison. Functoriality is obtained differently from the way it is formulated there. We never choose harmonic representatives compatibly across the filtration, which need not be possible since harmonic representatives rotate;

instead, exact fillings identify each homology group with a quotient of the fixed register cycle space, and the induced map is then forced to be the natural quotient.

The third layer is an exact source. A three-bit maximally mixed label turns a $\text{BQP}_1^{G_2}$ verifier with acceptance probability p_x into a verifier whose compressed acceptance operator on the label is $\text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0)$. Its perfect eigenspace has dimension 6 if $p_x = 1$ and 1 if $p_x \leq 1/3$, which fixes $\beta_d(X_{\text{in}}) = 8$ and the values $3/4$ and $1/8$. Finally, the common-copy blow-up [11], applied with the same labeled copy blocks at both levels, transfers Betti numbers, persistent rank, and gaps to unweighted graphs.

Technical overview.

Four ideas carry the proof of Theorem 1.2.

Finite certificates and the zero-weight limit. In normalized coordinates the boundary pair of a gadget with private weight λ is affine in λ , $D(\lambda) = D_0 + \lambda D_1$, and D_0 is a fixed rational matrix. The certificate (Assumption 3.1) asks for four exact facts about each palette member: the boundary intersection $B_{d_a}(Y_a) \cap V_a$ equals the range of the intended rank-one projector (exact filling); $\ker D_0 = V_a \oplus Q_a$ for an explicit private subspace Q_a ; a projected private differential is coercive on Q_a at first order in λ ; and its retained outputs are private. All four are integer or modular linear algebra, and Appendix C records them for the palette of this paper.

Harmonic padding. A certificate speaks about a constant-size active register, while the actual gadget is joined with the unaffected register factors, so the padded complex is exponentially large. The join Laplacian is a tensor sum over bidegrees, and the unaffected factors have harmonics only in top degree. We keep the local private data tensored with these outside harmonics and place every nonharmonic outside mode in a sector with a constant gap (Lemma 3.2). This is the step where the constants become independent of the padded dimension. A bulk consisting of all coordinates that contain a private vertex would not do, because deleting a weight-one outside vertex stays in that bulk with coefficient one.

Two energies. The proof of Equation (2) separates leakage out of the register from logical energy. Leakage is controlled by the zero-weight floor, the private coercivity, and an interface bound $\| [L_1 \cdots L_t] \|^2 = \| \sum_j L_j L_j^* \| \leq Ct\lambda^2$ that needs no orthogonality between the register outputs of different gadgets. Logical energy is controlled by exact filling through $\Delta^\dagger \succeq c\lambda^\kappa H^{\text{emb}}$. The two contributions carry different powers of λ , which is what decouples λ from g .

From estimate to exact multiplicity. Every vector of the spectral subspace of Δ below E projects nontrivially onto K , so that subspace has dimension at most $\dim K$. Exact fillings inject V/W_A into $H_d(X_A)$, providing $\dim K$ zero modes. Equality forces both the exact multiplicity and the absence of eigenvalues in $(0, E)$; no relative-acyclicity assumption is needed.

Scope.

The hardness is gate-dependent, as in the known perfect-completeness hardness results for topological data analysis [5, 15, 9]. We do not obtain ordinary BQP-hardness, unrestricted SDQC₁-hardness, or gate-independent BQP₁-hardness, and complex phases and general integer states are outside the certified palette. The initial Betti number of our instances is the constant 8, produced by three mixed label qubits. The gap is a complexity promise rather than a practical estimate, and the common-copy blow-up is polynomial but large. Section 8 discusses these points and the open problems they suggest.

Organization.

Section 2 fixes conventions and the problem. Section 3 states the certificate model and the transfer theorem, Section 4 derives quotient naturality, Section 5 instantiates the history construction with a certified palette and proves Theorem 5.3, and Section 6 transfers the result to unweighted graphs. Section 7 compares with prior work and Section 8 concludes. The appendices contain the complete transfer proof, the history-gap and source calculations, and the finite palette certificates.

2 Preliminaries and problem statement

Weighted clique complexes.

All chain spaces are finite-dimensional over $\mathbb{C}$. A vertex-weighted simplicial complex has positive vertex weights $w(v)$, and a simplex has weight $w(\sigma) = \prod_{v \in \sigma} w(v)$. We use the normalized coordinates of King and Kohler [15], in which the weighted boundary of an oriented simplex removes a vertex v with coefficient $\pm w(v)$; in matrix form $\partial_{w,k} = W_{k-1}^{-1} \partial_{1,k} W_k$, where W_k is the diagonal matrix of k -simplex weights. The weighted complex is therefore conjugate to the unweighted one and has the same homology. We use

$$\Delta_{X,d} = \partial_{X,d}^* \partial_{X,d} + \partial_{X,d+1} \partial_{X,d+1}^*, \quad Z_d(X) = \ker \partial_{X,d}, \quad B_d(X) = \text{ran } \partial_{X,d+1}.$$

Finite-dimensional Hodge theory [6, 12] gives the orthogonal decomposition $C_d(X) = \mathcal{H}_d(X) \oplus B_d(X) \oplus \text{ran } \partial_{X,d}^*$ with $\mathcal{H}_d(X) = \ker \Delta_{X,d}$ and $Z_d(X) = \mathcal{H}_d(X) \oplus B_d(X)$, and identifies $H_d(X) = Z_d(X)/B_d(X)$ with the harmonic space $\mathcal{H}_d(X)$. For a positive semidefinite operator A , write $\gamma_+(A) = \min(\text{spec}(A) \setminus \{0\})$ for its smallest positive eigenvalue, with $\gamma_+(0) = +\infty$. All certified boundary matrices and fillings below are rational. Rank, kernel dimension, image intersection, and filling validity are unchanged under scalar extension $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, and the real positive-semidefinite inequalities complexify verbatim. We state homology over $\mathbb{C}$ while certifying the finite palette over $\mathbb{Q}$.

185 Normalized persistence.

186 For $X_1 \subseteq X_2$, define

$$187 \quad \beta_d(X_1 \rightarrow X_2) = \text{rank}[H_d(X_1) \rightarrow H_d(X_2)], \quad q_d(X_1, X_2) = \frac{\beta_d(X_1 \rightarrow X_2)}{\beta_d(X_1)}. \quad (3)$$

188 Lowe et al. [19] write the numerator through harmonic projectors, as the rank of
 189 $P_{\mathcal{H}_d(X_2)}$ restricted to $\mathcal{H}_d(X_1)$. The two formulations agree. For a cycle $z \in Z_d(X_2)$, the
 190 vector $P_{\mathcal{H}_d(X_2)}z$ is the harmonic representative of $[z]$, because $z - P_{\mathcal{H}_d(X_2)}z \in B_d(X_2)$;
 191 hence $z \mapsto P_{\mathcal{H}_d(X_2)}z$ induces the isomorphism $H_d(X_2) \cong \mathcal{H}_d(X_2)$. Since $\mathcal{H}_d(X_1) \subseteq$
 192 $Z_d(X_1) \subseteq Z_d(X_2)$ and $\mathcal{H}_d(X_1) \cong H_d(X_1)$, the operator $P_{\mathcal{H}_d(X_2)}|_{\mathcal{H}_d(X_1)}$ is the induced
 193 map $H_d(X_1) \rightarrow H_d(X_2)$ up to these identifications, and its rank is $\beta_d(X_1 \rightarrow X_2)$.

194 ► **Definition 2.1** (Gapped normalized persistence). *An instance of the weighted problem*
 195 *consists of explicit simple graphs $G_1 \subseteq G_2$ on one labeled vertex set V , one common*
 196 *positive dyadic weight function $w : V \rightarrow \mathbb{Q}_{>0}$, a degree $d \geq 1$, positive rational lower*
 197 *bounds γ_1, γ_2 , and a positive rational accuracy ε , all encoded in binary and all at least*
 198 *$1/\text{poly}(|V|)$. Put $X_i = \text{Cl}(G_i)$, using the weighted-boundary convention above. The*
 199 *promises are*

$$200 \quad \beta_d(X_1) > 0, \quad \text{spec}(\Delta_{X_i, d}) \subseteq \{0\} \cup [\gamma_i, \infty) \quad (i = 1, 2).$$

201 *The task is to output $\tilde{q}$ such that $\Pr[|\tilde{q} - q_d(X_1, X_2)| \leq \varepsilon] \geq 2/3$. Input size is the*
 202 *total bit length of the graph encodings, weights, d , the two gap bounds, and ε . The*
 203 *unweighted problem is the restriction $w \equiv 1$. Its fixed decision version distinguishes*
 204 *$q_d \geq 3/4$ from $q_d \leq 1/8$.*

205 This is Problem 9 of Lowe et al. [19], with the same promises, including their inverse-
 206 polynomial lower bounds on the gap and accuracy parameters, and with the encoding
 207 conventions fixed. The denominator is the initial Betti number, not the number of
 208 simplices, and the statistic counts exact zero modes rather than eigenvalues below a
 209 cutoff.

210 Register, logical projectors, and gadgets.

211 The construction starts from a top-dimensional *register* complex R , so $C_{d+1}(R) = 0$,
 212 and sets $V = Z_d(R) = \mathcal{H}_d(R)$. Concretely, R is the clique complex of the join of
 213 n *bowtie* graphs, one per logical qubit. A bowtie is the wedge of two 4-cycles; its
 214 clique complex is one-dimensional and connected with $\beta_1 = 2$, and the two normalized
 215 oriented petal cycles are the computational basis $|0\rangle, |1\rangle$. By the Künneth formula for
 216 joins, R has reduced homology only in degree $d = 2n - 1$, and $V \cong (\mathbb{C}^2)^{\otimes n}$ isometrically.
 217 Each constraint j is a rank-one projector π_j on the cycle space of a constant-size *active*
 218 sub-register; after tensoring with the identity of the unaffected factors it becomes an
 219 orthogonal projector Π_j on V , generally of higher rank. For a term set A , put

$$220 \quad H_A = \sum_{j \in A} \Pi_j, \quad K_A = \ker H_A, \quad W_A = \sum_{j \in A} \text{ran } \Pi_j = K_A^\perp, \quad (4)$$

221 and use a certified logical gap

$$222 \quad H_A \succeq g_A(P_V - P_{K_A}), \quad 0 < g_A \leq 1. \quad (5)$$

223 Each constraint is realized by a weighted clique gadget: a graph containing the
 224 active register, whose additional *private* vertices have weight λ and are joined to
 225 every vertex of the unaffected register factors, while all register vertices have weight
 226 one. Attaching the gadgets of all $j \in A$ to R gives the clique complex X_A , and Δ_A
 227 denotes its full Laplacian on all degree- d chains. Thus H_A acts on V , while Δ_A acts
 228 on $C_d(X_A) \supseteq V$.

229 ► **Definition 2.2** (Independent private interiors). *The gadgets of a term set A have
 230 independent private interiors if private vertices of distinct gadgets are never adjacent.
 231 Every simplex of X_A that contains a private vertex then contains private vertices of
 232 exactly one gadget, and $C_d(X_A) = C_d(R) \oplus \bigoplus_{j \in A} C_{d,j}^{\text{priv}}$ as coordinate spaces.*

233 All constructions in this paper satisfy Definition 2.2: distinct gadgets receive disjoint
 234 sets of new vertices, and no edge is added between private vertices of distinct gadgets.

235 Circuit source.

236 For the circuit application, J inserts the fixed clean string around a maximally mixed
 237 logical input, U is an exact circuit, and

$$238 \quad M = J^* U^* P_{\text{acc}} U J, \quad S = \ker(I - M).$$

239 The initial history kernel will have the full input dimension, while the final history
 240 kernel will be isomorphic to the perfect accepting space S .

241 3 Finite-certificate transfer to the complete geometric kernel

242 We isolate finite local conditions under which a clique-gadget construction controls
 243 every chain of the full geometric complex, including chains supported mostly outside
 244 the simulated register. The statement does not use convergence of a spectral projector
 245 whose rank grows with the simulated circuit.

246 3.1 The certificate model

247 Consider one active gadget before padding. Its weight-one register is R_a , its target
 248 cycle space is $V_a = Z_{d_a}(R_a)$, and every private vertex has weight λ . In normalized
 249 chain coordinates put

$$250 \quad D_a(\lambda) = \begin{bmatrix} \partial_{d_a}(\lambda) \\ \partial_{d_a+1}(\lambda)^* \end{bmatrix} = D_{a,0} + \lambda D_{a,1},$$

251 so that $\Delta_{Y_a, d_a}(\lambda) = D_a(\lambda)^* D_a(\lambda)$. The decomposition is affine in λ because every
 252 entry of the normalized boundary is 0, ± 1 (a register vertex removed) or $\pm \lambda$ (a private
 253 vertex removed). The value at $\lambda = 0$ is a matrix limit, not a singular change of basis;
 254 $D_{a,0}$ is the boundary pair of a rational chain complex, since $\partial(\lambda)^2 = 0$ for all λ forces
 255 $\partial_0^2 = 0$.

► **Assumption 3.1** (Finite local certificate). *For every member a of a fixed finite palette there are an orthogonal subspace Q_a contained in the span of private simplex coordinates, an output projector R_a^{out} , and a rank-one projector π_a on V_a such that:*

- (i) $B_{d_a}(Y_a) \cap V_a = \text{ran } \pi_a$ for every $\lambda > 0$;
- (ii) $\ker D_{a,0} = V_a \oplus Q_a$, orthogonally;
- (iii) for $T_a(\lambda) = R_a^{\text{out}} D_a(\lambda)$,

$$T_a(\lambda) = \lambda T_{a,1}, \quad T_{a,1} V_a = 0, \quad \|T_{a,1} q\| \geq b \|q\| \quad (q \in Q_a);$$

- (iv) every retained output coordinate contains a private vertex.

The constant $b > 0$, the positive singular-value floor of $D_{a,0}$, and all matrix norms are uniform over the palette.

All data in Assumption 3.1 are finite integer or rational linear algebra. Exact fillings prove one inclusion in (i); matching boundary ranks and surviving register cosets, or the guard connecting-map argument, prove the reverse inclusion. Condition (i) does not depend on λ : the weighted boundary space is $W_{d_a}^{-1} B_{d_a}(Y_a; 1)$ and W_{d_a} is the identity on register coordinates. Modular rank bounds with matching rational kernels certify (ii), and a nonzero Gram determinant certifies (iii). The subspace Q_a need not be spanned by simplex coordinates; the selected-cycle guards of Appendix C produce Q_a of the form $Q \otimes \mathbb{C}\gamma$. The explicit palette used later is recorded in Appendix C.

► **Lemma 3.2** (Weight scaling and harmonic padding). *Let $0 < \lambda \leq 1$, let $d_a = 2m_a - 1$ with $m_a \leq m$, and put $\kappa := 4m + 2$. There is a palette constant $c > 0$ such that*

$$\gamma_+(\Delta_{Y_a, d_a}(\lambda)) \geq c \lambda^\kappa. \quad (6)$$

Let Y_a be joined with the unaffected register factors, whose join R_{out} has reduced homology only in its top degree, with top harmonic space $\mathcal{H}_{\text{out}}$, and let d be the global target degree. Then the zero-weight kernel of the padded gadget in degree d is

$$V \oplus (Q_a \otimes \mathcal{H}_{\text{out}}),$$

its orthogonal complement has a uniform positive zero-weight floor, the projected pair $T_a(\lambda) \otimes P_{\mathcal{H}_{\text{out}}}$ satisfies (iii) with the same constant b , and the padded exact boundary intersection with V is the range of $\Pi_a = \pi_a \otimes I_{\mathcal{H}_{\text{out}}}$.

Proof. See Appendix A.1. ◀

The harmonic restriction in this lemma is essential. If the padded bulk contained every coordinate with a local private vertex, deleting a weight-one outside vertex could remain in that bulk with coefficient one. We instead keep local private coordinates tensored with outside harmonics and place every nonharmonic outside factor in the separately gapped sector.

3.2 Global theorem

Attach t_A padded gadgets to the common top-dimensional register R with independent private interiors (Definition 2.2). Let L_j be the block of $\partial_{X_A, d}$ that maps the private

293 d -chains of gadget j to the register $(d-1)$ -chains. A private d -simplex has a register
 294 face only if it contains exactly one private vertex, and removing that vertex gives
 295 coefficient $\pm\lambda$; hence every column of $\lambda^{-1}L_j$ has at most one nonzero entry and every
 296 row has at most p_j nonzero entries, where p_j is the number of private vertices of gadget
 297 j . Consequently the finite interface bound

$$298 \quad L_j L_j^* \preceq N_j \lambda^2 I, \quad N_j \leq N_*, \quad (7)$$

299 holds with $N_j \leq p_j$, so N_* is a palette constant. The input blocks are orthogonal,
 300 while the outputs may overlap. Hence, for $L = [L_1 \ \cdots \ L_{t_A}]$,

$$301 \quad LL^* = \sum_j L_j L_j^* \preceq N_* t_A \lambda^2 I.$$

302 **► Theorem 3.3** (Finite-certificate degenerate-kernel transfer). *Fix a finite palette sat-*
 303 *isfying Assumption 3.1, with active degrees $d_a = 2m_a - 1$, $m_a \leq m$, and let $\kappa = 4m + 2$.*
 304 *Let $\mathcal{A}$ be a finite family of term sets whose gadgets are padded palette members attached*
 305 *to the register R of Section 2 with independent private interiors, all with the same*
 306 *private weight λ . Let $t_{\max} = \max\{1, \max_{A \in \mathcal{A}} |A|\}$ and suppose*

$$307 \quad H_A = \sum_{j \in A} \Pi_j \succeq g_A (P_V - P_{K_A}), \quad K_A = \ker H_A, \quad 0 < g_A \leq 1$$

308 *for every $A \in \mathcal{A}$. There are palette constants $c_0, C > 0$ such that, whenever $0 < \lambda \leq$*
 309 *$\min\{1, c_0/t_{\max}\}$, every unit $x \in C_d(X_A)$ satisfies*

$$310 \quad \|(I - P_{K_A})x\|^2 \leq C \left[t_A \lambda^2 + \frac{\langle x, \Delta_A x \rangle}{g_A \lambda^\kappa} \right]. \quad (8)$$

311 *Here P_{K_A} is the projector onto the embedded logical kernel, extended by zero outside*
 312 *the register.*

313 *For $0 < \eta < 1$, choose additionally*

$$314 \quad \lambda \leq c_1 \min\{t_{\max}^{-1}, \eta/\sqrt{t_{\max}}\}, \quad E_A \leq c_2 \eta^2 g_A \lambda^\kappa, \quad (9)$$

315 *with palette constants $c_1, c_2 > 0$. Then*

$$316 \quad \dim \ker \Delta_A = \dim K_A, \quad \text{spec}(\Delta_A) \cap (0, E_A) = \emptyset. \quad (10)$$

317 *This includes $K_A = 0$, in which case $\Delta_A \succeq E_A I$.*

318 **Proof architecture.**

319 The full calculation is in Appendix A. The shared-output identity above and the
 320 private projected pairs first give the geometric leakage estimate

$$321 \quad \|(I - P_V)x\|^2 \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2}) \langle x, \Delta_A x \rangle \right]. \quad (11)$$

322 Exact filling and Lemma 3.2 separately give

$$323 \quad \Delta_A^\uparrow \succeq c \lambda^\kappa H_A^{\text{emb}}, \quad (12)$$

where $\Delta_A^\uparrow = \partial_{d+1} \partial_{d+1}^*$ and H_A^{emb} is H_A extended by zero outside V . Combining these inequalities with the logical gap proves Equation (8). The bound applies to every vector in a low-energy spectral subspace. Projection onto K_A is therefore injective on that subspace, so its dimension is at most $\dim K_A$. Independent exact fillings supply $\dim K_A$ zero modes through V/W_A ; equality and the whole-positive-spectrum gap follow. No relative-acyclicity assumption is used.

For the explicit history palette $m \leq 6$, so $\kappa = 26$. Taking a dyadic $\lambda = \Theta(\eta/t_{\max})$ gives the conservative scale

$$E_A = \Omega\left(\frac{\eta^{28} g_A}{t_{\max}^{26}}\right). \quad (13)$$

The exponent is not claimed to be optimal.

4 Filtered quotient naturality

The transfer theorem determines the dimension of every endpoint kernel. Exact filling determines the maps between those kernels at the homology level. This separation is useful: spectral concentration proves that there are no extra classes, while the quotient model computes persistence without comparing harmonic representatives.

► **Lemma 4.1** (Independent filling). *For every term set A ,*

$$B_d(X_A) \cap V = W_A = \sum_{j \in A} \text{ran } \Pi_j. \quad (14)$$

Proof. Every vector in $\text{ran } \Pi_j$ has a filling in gadget j , so $W_A \subseteq B_d(X_A) \cap V$. Conversely, write a degree- $(d+1)$ chain as a sum of its gadget components. If its total boundary lies in V , then its boundary has zero coordinate on every private d -simplex. Private supports of distinct gadgets are disjoint, so this vanishing holds separately for each gadget component. The remaining register boundary of component j lies in $B_d(Y_j) \cap V = \text{ran } \Pi_j$. Summing over j proves the reverse inclusion. ◀

No relative-acyclicity hypothesis appears in Lemma 4.1. Extra geometric homology outside V is excluded later by the all-chain concentration theorem, not by the boundary decomposition alone.

► **Theorem 4.2** (Natural quotient realization). *Under the hypotheses and parameter choice of Theorem 3.3, the map*

$$q_A : V/W_A \longrightarrow H_d(X_A), \quad q_A(v + W_A) = [v], \quad (15)$$

is an isomorphism. For $A \subseteq B$, these isomorphisms make the diagram

$$\begin{array}{ccc} V/W_A & \xrightarrow{\rho_{AB}} & V/W_B \\ q_A \downarrow \cong & & \downarrow \cong q_B \\ H_d(X_A) & \xrightarrow{(\iota_{AB})_*} & H_d(X_B) \end{array}$$

commute, where ρ_{AB} is the natural quotient map. Consequently,

$$\text{rank}(\iota_{AB})_* = \dim(V/W_B) = \dim K_B. \quad (16)$$

357 **Proof.** See Appendix A.6. ◀

358 For a chain $A_0 \subseteq A_1 \subseteq \dots \subseteq A_s$, every arrow in degree d is an epimorphism. The
 359 constructed barcode therefore has deaths but no new births in this degree. More
 360 generally, for any $i \leq j$,

$$361 \quad \text{rank}[H_d(X_{A_i}) \rightarrow H_d(X_{A_j})] = \dim K_{A_j}.$$

362 Harmonic representatives may rotate between filtration levels; the exact statement
 363 concerns the algebraic homology maps.

364 Taking A_{in} to be the clock, propagation, and clean-input terms of a history
 365 Hamiltonian, and taking A_{out} to add the final rejection term, gives

$$366 \quad q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim K_{\text{out}}}{\dim K_{\text{in}}}. \quad (17)$$

367 This identity is the bridge from exact logical perfect-space fractions to normalized
 368 persistence.

369 **5 Exact histories, a fixed clique palette, and hardness**

370 Let

$$371 \quad G_2 = \{X, CX, CCX, H \otimes H\}.$$

372 Following Rudolph's gate-dependent definition [22, Definition 2.2],¹ $\text{BQP}_1^{G_2}$ consists of
 373 promise problems decided by a polynomial-time uniform family of exact G_2 circuits
 374 on an all-zero input, with one computational-basis output bit, perfect completeness,
 375 and inverse-polynomial rejection probability in the no case. Rudolph's exact error
 376 reduction [22, Theorem 2.3] preserves perfect completeness and stays in any gate set
 377 containing G_2 ; we apply it first so that soundness is at most $1/3$. The superscript
 378 is essential: no approximate compilation or gate-independent identification is used
 379 below.

380 **5.1 Rational histories and the certified palette**

381 Write $K = \sqrt{2}H$. Replace each $H \otimes H$ transition by a gain edge K on one work
 382 bit followed by a loss edge $K^{-1} = H/\sqrt{2}$ on the other. Their composite is exactly
 383 $H \otimes H$, and each edge is enforced by two orthonormal rational three-term vectors.
 384 The vectors and their direct calculation appear in Appendix B.1. This construction is
 385 due to Rudolph [22, Appendix D.1].

386 The propagation constraints for X, CX, CCX decompose, on their constant-size
 387 active support, into two-time vectors

$$388 \quad (|0, z\rangle - |1, Uz\rangle)/\sqrt{2}.$$

¹ Definition, theorem, and appendix numbers for [22] refer to the full version, arXiv:2411.02681v2.

Clock, input, and output penalties are basis projectors. Unaffected work qubits are identities and are implemented by the harmonic padding of Lemma 3.2; they are not expanded into exponentially many rank-one terms.

► **Lemma 5.1** (Certified fixed palette). *Every local constraint in the split G_2 history is implemented by a member of one fixed family of weighted clique gadgets of total support locality at most six. Every member satisfies Assumption 3.1 and Equation (7), with uniform constants, and the graph construction is polynomial in the circuit size.*

Proof. See Appendix C. It records the finite integer certificates, exact relabelings, selected-cycle guard closure, and polynomial-size bookkeeping. ◀

5.2 Complete history kernels

Let L be the number of legal clock positions after splitting the Hadamard transitions. Put $s_t = \sqrt{2}$ immediately after a gain edge and $s_t = 1$ otherwise. If V_t is the unitary prefix after removing these scalars, then

$$\mathcal{V}v = Z^{-1/2} \sum_{t=0}^{L-1} s_t |t\rangle V_t v, \quad Z = \sum_t s_t^2, \quad L \leq Z \leq 2L. \quad (18)$$

Every gain edge is immediately followed by its matching loss edge, with no gate between them, so $s_0 = s_{L-1} = 1$.

► **Proposition 5.2** (History kernels and gaps). *Let C be the valid clean-input space, $D = \dim C > 0$, and let $M = J^* U^* P_{\text{acc}} U J$ be the compressed acceptance operator. Put $S = \ker(I - M)$. The initial history Hamiltonian has complete kernel $\mathcal{V}C$, while adding the final rejection projector gives complete kernel $\mathcal{V}S$. If $M|_{S^\perp} \preceq I/3$, then*

$$g_{\text{in}} \geq \frac{1}{8L^2}, \quad g_{\text{out}} \geq \frac{1}{3L(8L^2 + 1)} \geq \frac{1}{27L^3}. \quad (19)$$

The second statement includes $S = 0$.

The proof in Appendix B uses a weighted path Poincaré inequality. The output compression is exactly

$$\mathcal{V}^* H_{\text{rej}} \mathcal{V} = (I - M)/Z, \quad (20)$$

and a two-positive-operator estimate turns this angle penalty into the second gap in Equation (19). Combining Theorem 3.3, Theorem 4.2, Lemma 5.1, and Proposition 5.2 yields weighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ with

$$\beta_d(X_{\text{in}}) = D, \quad \beta_d(X_{\text{out}}) = \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \dim S, \quad (21)$$

and inverse-polynomial gaps above both complete kernels.

5.3 A fixed-eight source

Given a $\text{BQP}_1^{G_2}$ verifier Q_x with acceptance probability p_x , add a maximally mixed label $z \in \{0, 1\}^3$, run Q_x unconditionally, and accept according to

$$[z = 000] \vee (q \wedge [z \in \{001, 010, 011, 100, 101\}]),$$

where q is the output of Q_x . The reversible predicate uses only X, CX, CCX , preserves z , and uncomputes its scratch. Its compressed acceptance operator on the label is exactly

$$M_x = \text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0). \quad (22)$$

Thus the perfect eigenspace has dimension six when $p_x = 1$ and dimension one when $p_x \leq 1/3$; every off-perfect eigenvalue in the latter case is at most $1/3$.

► **Theorem 5.3** (Fixed-eight hardness). *There is a deterministic polynomial-time many-one reduction from $\text{BQP}_1^{G_2}$ to the decision version of weighted gapped normalized persistence (Definition 2.1). Every output satisfies*

$$\beta_d(X_{\text{in}}) = 8, \quad \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \begin{cases} 6, & x \in L, \\ 1, & x \notin L, \end{cases}$$

and both endpoint degree- d Laplacians have inverse-polynomial gaps above their complete kernels. Hence the normalized values are $3/4$ and $1/8$. Additive error $1/24$, with decision threshold $7/16$, distinguishes the two cases with success probability at least $2/3$.

Proof. Apply Equation (21) to Equation (22). The two ratios differ by $5/8$, and $1/24 < 5/16$. The logical gaps in Equation (19) are explicit rationals. The fixed palette supplies hard-coded positive rational lower bounds for the constants in Theorem 3.3. Fix $\eta_0 = 1/10$; from the polynomially bounded t_{max} , choose the first integer b for which $\lambda = 2^{-b}$ satisfies both inequalities in Equation (9). For each endpoint let s_i be the least integer with $2^{-s_i} \leq \frac{1}{2}c_2\eta_0^2g_i\lambda^{26}$, and output $E_i = 2^{-s_i}$, which satisfies Equation (9). Thus $b = O(\log t_{\text{max}})$ and $s_i = O(\log t_{\text{max}} + \log L)$, so the binary encodings of λ and E_i have polynomial length. If the history uses n register qubits (clock and work bits), the register has $7n$ vertices and $O(n^2)$ edges, and each gadget adds a constant number of private vertices together with $O(n)$ edges to the unaffected register factors; so the weighted graph has $O(n + t_{\text{max}})$ vertices and $O(n^2 + t_{\text{max}}n)$ explicitly generated edges. Hence the map outputs the two graphs, their common weight function, d, E_1, E_2 , and the fixed decision threshold in deterministic polynomial time. ◀

► **Remark 5.4** (Integer weights). Multiplying every vertex weight by 2^b gives an equivalent instance with integer weights in $\{1, 2^b\}$: the normalized boundary $W_{k-1}^{-1}\partial W_k$ scales by 2^b , so the Laplacian and both gap bounds scale by 2^{2b} , while homology is unchanged. The weighted instances therefore also satisfy the polynomially bounded weight convention of [19].

► Remark 5.5 (Other gate sets). By Rudolph’s exact simulation theorem [22, Theorem 3.4], $\text{BQP}_1^G \subseteq \text{BQP}_1^{G_2}$ for every finite gate set G whose entries lie in a cyclotomic field $\mathbb{Q}(\zeta_{2^k})$, $k \in \mathbb{N}$; Clifford+ T is an example. Composing that simulation with Theorem 5.3 gives BQP_1^G -hardness for each such G separately. This is a gate-family-relative statement, not gate-independent BQP_1 -hardness.

The exact trace-versus-perfect-space obstruction and the mixed-spectator extension to individual real Hadamards are recorded in Appendix B.5. Neither is needed for Theorem 5.3.

6 A natural unweighted corollary

The weighted construction uses only weights 1 and a dyadic inverse-polynomial λ . Hayakawa’s common-copy construction replaces each vertex v by a clique of f_v labeled copies and each base edge by all cross-block edges [11]. Choose one common $F = \lambda^{-2}$ and

$$f_v = Fw(v)^2.$$

Register vertices then have F copies and private vertices have one.

In degree k , let $U_{i,k}$ map a normalized base simplex to the normalized average of its one-copy-per-block lifts. Lemma 4.2 and Theorem 4.3 of Hayakawa [11] give the symmetric identity and asymmetric gap over real chains; scalar extension gives the same statements over $\mathbb{C}$:

$$\widehat{\Delta}_{i,k} U_{i,k} = F U_{i,k} \Delta_{i,k}, \quad \widehat{\Delta}_{i,k}|_{\text{ran}(U_{i,k})^\perp} \succeq \min_v f_v I. \quad (23)$$

Use the same labeled copy block for every shared vertex at both filtration levels. With consistent orientations, simplicial inclusion then satisfies

$$\widehat{J}_{ij} U_{i,d} = U_{j,d} J_{ij}. \quad (24)$$

There is no extra normalization factor because an inclusion preserves the simplex degree, weights, labels, and copy set.

► **Corollary 6.1** (Unweighted normalized persistence). *The reduction in Theorem 5.3 maps in polynomial time to nested unweighted clique complexes with the same endpoint Betti numbers, persistent rank, and normalized values $3/4$ and $1/8$. Both unweighted endpoint Laplacians retain inverse-polynomial gaps above their complete kernels.*

Proof. The asymmetric bound in Equation (23) excludes asymmetric harmonic vectors, so

$$\ker \widehat{\Delta}_{i,d} = U_{i,d} \ker \Delta_{i,d}.$$

Together with Equation (24), this makes every inclusion-induced map on harmonic spaces unitarily equivalent to its weighted counterpart. Endpoint nullities and induced ranks are therefore preserved. If E_i lower-bounds the positive weighted spectrum, then

$$\gamma_+(\widehat{\Delta}_{i,d}) \geq \min\{F E_i, \min_v f_v\} = \min\{F E_i, 1\}.$$

By Equation (13), $FE_i = \Omega(\eta^{26}g_i/t_{\max}^{24})$. Since F is polynomial, so is the blown-up graph. More explicitly, if the weighted graph has $O(n + t_{\max})$ vertices and $O(n^2 + t_{\max}n)$ edges, a safe explicit bound after blow-up is $O(F(n + t_{\max}))$ vertices and $O(F^2(n^2 + t_{\max}n))$ edges; labels and edges are generated by nested loops in polynomial time. Future private vertices may be present initially as isolated vertices; the target degree is positive, so they add no target-degree chain. ◀

The operator decomposition is Hayakawa's; the filtration-specific step is the commuting identity Equation (24). We do not count the unweighting mechanism as a separate contribution.

7 Relation to prior work

Normalized persistence.

Lowe et al. [19] define normalized harmonic persistence (their Problem 9, our Definition 2.1), prove that a low-energy relaxation of it is DQC_1 -hard and in BQP, prove SDQC_1 -hardness of the local-Hamiltonian analogue, and conjecture SDQC_1 -hardness of Problem 9 (their Conjecture 1). Their Conjecture 2 asks for a polynomial-time map from history Hamiltonians H_1 and $H_2 = H_1 + \lambda H_{\text{out}}$ to weighted clique complexes $X_1 \subseteq X_2$ with two properties: kernel fidelity and gap, $\beta_d(X_i) = \dim \ker H_i$ together with computable inverse-polynomial gap bounds; and functoriality, $\beta_d(X_1 \rightarrow X_2) = \dim \text{ran}(P_{\ker H_2}|_{\ker H_1})$. Their Lemma 11 shows that Conjecture 2 implies Conjecture 1. Theorems 3.3 and 4.2 establish both properties for the exact G_2 history Hamiltonians of Section 5, with H_{out} at unit coefficient rather than at a small coefficient λ , and with the induced map computed through the quotient model rather than through compatible harmonic isometries as formulated there. Two differences prevent us from concluding Conjecture 1 itself. First, our source is $\text{BQP}_1^{G_2}$ with three mixed label qubits rather than an SDQC_1 verifier with a maximally mixed register. The SDQC_1 promise constrains the acceptance probability of the maximally mixed state, and a trace promise does not in general determine the dimension of the perfectly accepted subspace, which is what q_d measures (Appendix B.5). Second, the certified palette is real and exact, so verifiers over an arbitrary gate set are not covered. Their containment result (Lemma 7) needs additional access assumptions, namely a preparation circuit for the uniform mixture over $\ker \Delta_{1,d}$ and an inverse-polynomial lower bound on the nonzero principal angles between the two harmonic spaces. We do not establish these for our instances, and containment of the promise problem of Definition 2.1 in BQP remains open.

Clique homology gadgets.

Crichigno and Kohler [5] give a parsimonious reduction in which satisfying states correspond one-to-one to holes, so exact multiplicity by itself is not new; their instances carry no gap promise. King and Kohler [15] introduce the weighted gadgets and the many-gadget spectral analysis behind gapped clique homology, proving a gap when the logical Hamiltonian is positive definite, with a gadget weight proportional to the logical

gap; Hayakawa [11] removes the weights under the same positive-definite promise. Our transfer theorem is formulated from finite zero-weight certificates, applies to arbitrary geometric chains and to degenerate kernels, and decouples λ from g . Its padding step differs from the coordinate-bulk formulation displayed in the arXiv version of King and Kohler [14, Claim 10.2]: there the bulk consists of all coordinates containing a private vertex, which in the padded complex also contains boundary terms obtained by deleting a weight-one outside vertex, whereas we keep local private coordinates tensored with outside harmonics in every bidegree (Lemma 3.2). We also work with the fixed local matrices before tensoring with the outside identity, so that no operator bound on a projector of growing rank has to be inferred from basiswise perturbation estimates. These are differences of proof technique, and we make no claim about the presentation in the published version [15].

543 Persistence hardness.

Gyurik et al. [9] prove BQP_1 -hardness, over the gate set $\{\text{CNOT}, U_{\text{Pyth}}\}$, of deciding whether a *given* harmonic representative of X_1 survives in X_2 , with a gap promise on the final Laplacian. Normalized persistence asks instead for the fraction of the full initial homology that survives, with no representative given, so a hardness proof must control the complete kernels at both endpoints. Persistent Laplacians [21] give a spectral formulation of persistent Betti numbers; we use ordinary endpoint Laplacians for the promise and the quotient model for the map.

551 Sources and gate sets.

Rudolph [22] defines the gate-dependent classes, proves exact error reduction inside gate sets containing G_2 , shows $\text{BQP}_1^G \subseteq \text{BQP}_1^{G_2}$ for every finite G with entries in some $\mathbb{Q}(\zeta_{2^k})$, and supplies the gain/loss split of $H \otimes H$ with rational three-term constraint vectors and the base clique gadgets used in our palette. Cade and Crichigno [3] relate cohomology problems to QMA_1 and DQC_1 , and Gyurik, Cade, and Dunjko [8] connected quantum topological data analysis to DQC_1 [17]. The history-state construction with a unary clock and neighbor guards is Kitaev's [16, 13], with the rational amplitude bookkeeping of Section 5.

560 Unweighting.

Hayakawa's symmetric/asymmetric decomposition of the common-copy blow-up [11, Lemma 4.2 and Theorem 4.3] is used as a black box. Our only addition is the observation that identical labeled copy blocks at both filtration levels make the symmetric isometries commute with inclusion, Equation (24). We do not count this as a separate contribution.

566 8 Limitations and open problems

The construction has four substantive limits. Its exact palette is finite and real: arbitrary integer states and complex phases are outside the theorem, so the hardness source

is the gate-dependent class $\text{BQP}_1^{G_2}$ rather than ordinary BQP , unrestricted SDQC_1 , or a gate-independent perfect-completeness class. The clean source has $\beta_d(X_{\text{in}}) = 8$; ignored mixed qubits replicate this denominator exactly, but an intrinsically growing family of distinct initial classes would require another exact source. The gap is a complexity promise rather than a practical estimate: at locality six it scales conservatively as $\Omega(\eta^{28}g/t^{26})$ with $g = \Omega(L^{-3})$, and the common-copy blow-up multiplies the vertex count by $F = \lambda^{-2}$. Finally, the hardness is for the promise problem with endpoint gaps; whether our instances also satisfy the access assumptions under which Lowe et al. place the problem in BQP is not established.

Two questions would strengthen the complexity classification.

1. Is normalized harmonic persistence SDQC_1 -hard, as conjectured in [19]? The missing steps are a source whose perfect-space fraction is separated by the SDQC_1 trace promise, and a certified palette for the corresponding gate set.
2. Do the constructed filtrations satisfy an inverse-polynomial lower bound on the nonzero principal angles between $\mathcal{H}_d(X_{\text{in}})$ and $\mathcal{H}_d(X_{\text{out}})$, and admit efficient preparation of the initial harmonic mixture? Establishing both would place the instances in BQP by [19, Lemma 7].

Within its limits, the paper establishes a reusable criterion: a new clique reduction obtains exact kernel multiplicity, an inverse-polynomial gap above the whole kernel, and natural persistent ranks by certifying, for a finite palette, local zero-weight kernels, private coercivity, exact filling, and interfaces, with no growing-dimensional perturbation argument.

References

- 1 Simon Apers, Sander Gribling, Sayantan Sen, and Dániel Szabó. A (simple) classical algorithm for estimating Betti numbers. *Quantum*, 7:1202, 2023. doi:10.22331/q-2023-12-06-1202.
- 2 Dominic W. Berry, Yuan Su, Casper Gyurik, Robbie King, Joao Basso, Alexander Del Toro Barba, Abhishek Rajput, Nathan Wiebe, Vedran Dunjko, and Ryan Babbush. Analyzing prospects for quantum advantage in topological data analysis. *PRX Quantum*, 5:010319, 2024. doi:10.1103/PRXQuantum.5.010319.
- 3 Chris Cade and P. Marcos Crichigno. Complexity of supersymmetric systems and the cohomology problem. *Quantum*, 8:1325, 2024. doi:10.22331/q-2024-04-30-1325.
- 4 Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009. doi:10.1090/S0273-0979-09-01249-X.
- 5 Marcos Crichigno and Tamara Kohler. Clique homology is QMA₁-hard. *Nature Communications*, 15:9846, 2024. URL: <https://www.nature.com/articles/s41467-024-54118-z>, doi:10.1038/s41467-024-54118-z.
- 6 Beno Eckmann. Harmonische Funktionen und Randwertaufgaben in einem Komplex. *Commentarii Mathematici Helvetici*, 17:240–255, 1944. doi:10.1007/BF02566245.
- 7 Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002. doi:10.1007/s00454-002-2885-2.
- 8 Casper Gyurik, Chris Cade, and Vedran Dunjko. Towards quantum advantage via topological data analysis. *Quantum*, 6:855, 2022. doi:10.22331/q-2022-11-10-855.
- 9 Casper Gyurik, Alexander Schmidhuber, Robbie King, Vedran Dunjko, and Ryu Hayakawa. Provable quantum speedups for computing persistence in topological data analysis. *PRX Quantum*, 7:020361, 2026. URL: <https://arxiv.org/abs/2410.21258v2>, doi:10.1103/gvys-h18h.
- 10 Ryu Hayakawa. Quantum algorithm for persistent Betti numbers and topological data analysis. *Quantum*, 6:873, 2022. doi:10.22331/q-2022-12-07-873.
- 11 Ryu Hayakawa. Unweighted gapped clique homology is QMA₁-complete, 2026. URL: <https://arxiv.org/abs/2608.02726v1>, arXiv:2608.02726.
- 12 Danijela Horak and Jürgen Jost. Spectra of combinatorial Laplace operators on simplicial complexes. *Advances in Mathematics*, 244:303–336, 2013. doi:10.1016/j.aim.2013.05.007.
- 13 Julia Kempe, Alexei Kitaev, and Oded Regev. The complexity of the local Hamiltonian problem. *SIAM Journal on Computing*, 35(5):1070–1097, 2006. doi:10.1137/S0097539704445226.
- 14 Robbie King and Tamara Kohler. Gapped clique homology on weighted graphs is QMA₁-hard and contained in QMA, 2023. Version 2, equation-level proof comparison. URL: <https://arxiv.org/abs/2311.17234v2>, arXiv:2311.17234.

- 634 **15** Robbie King and Tamara Kohler. Gapped clique homology on weighted graphs
635 is QMA_1 -hard and contained in QMA . *SIAM Journal on Computing*, pages
636 FOCS24–137–FOCS24–235, 2026. doi:10.1137/24M1710243.
- 637 **16** Alexei Yu. Kitaev, Alexander H. Shen, and Mikhail N. Vyalı. *Classical and*
638 *Quantum Computation*, volume 47 of *Graduate Studies in Mathematics*. American
639 Mathematical Society, 2002.
- 640 **17** Emanuel Knill and Raymond Laflamme. Power of one bit of quantum information.
641 *Physical Review Letters*, 81(25):5672–5675, 1998. doi:10.1103/PhysRevLett.
642 81.5672.
- 643 **18** Seth Lloyd, Silvano Garnerone, and Paolo Zanardi. Quantum algorithms for
644 topological and geometric analysis of data. *Nature Communications*, 7:10138,
645 2016. doi:10.1038/ncomms10138.
- 646 **19** Dominic Lowe, M. S. Kim, Roberto Bondesan, and Ryu Hayakawa. Complexity of
647 normalized persistence problems for topological data analysis and local hamiltoni-
648 ans, 2026. URL: <https://arxiv.org/abs/2607.03278v1>, arXiv:2607.03278.
- 649 **20** Sam McArdle, András Gilyén, and Mario Berta. A streamlined quantum algorithm
650 for topological data analysis with exponentially fewer qubits. *Quantum*, 10:2058,
651 2026. doi:10.22331/q-2026-04-10-2058.
- 652 **21** Facundo Mémoli, Zhengchao Wan, and Yusu Wang. Persistent laplacians: Prop-
653 erties, algorithms and implications. *SIAM Journal on Mathematics of Data*
654 *Science*, 4(2):858–884, 2022. doi:10.1137/21M1435471.
- 655 **22** Dorian Rudolph. Towards a universal gateset for QMA_1 . In Michal Koucký
656 and Daniela Petrişan, editors, *51st International Symposium on Mathemat-*
657 *ical Foundations of Computer Science (MFCS 2026)*, volume 386 of *Leibniz*
658 *International Proceedings in Informatics (LIPIcs)*, pages 98:1–98:19. Schloss Dag-
659 stuhl – Leibniz-Zentrum für Informatik, 2026. Full version: arXiv:2411.02681v2.
660 doi:10.4230/LIPIcs.MFCS.2026.98.
- 661 **23** Alexander Schmidhuber and Seth Lloyd. Complexity-theoretic limitations on
662 quantum algorithms for topological data analysis. *PRX Quantum*, 4:040349, 2023.
663 doi:10.1103/PRXQuantum.4.040349.
- 664 **24** Afra Zomorodian and Gunnar Carlsson. Computing persistent homology. *Discrete*
665 *& Computational Geometry*, 33:249–274, 2005. doi:10.1007/s00454-004-114
666 6-y.

667 **A** Proof of the all-chain transfer theorem

668 This appendix proves Theorem 3.3 from the finite local hypotheses. All constants
669 below depend only on the fixed palette.

670 **A.1** Local scaling and harmonic padding

671 Let W_k be the diagonal matrix of simplex weights. In normalized coordinates,

$$672 \quad \partial_{w,k} = W_{k-1}^{-1} \partial_{1,k} W_k.$$

673 For $0 < \lambda \leq 1$, invertible left and right scaling at the fixed rank of $\partial_{1,k}$ gives

$$674 \quad \sigma_{\min}^+(\partial_{w,k}) \geq \lambda^{k+1} \sigma_{\min}^+(\partial_{1,k}).$$

675 Indeed, put $K = \ker \partial_{1,k}$. For every unit $x \perp W_k^{-1}K$,

$$676 \quad \text{dist}(W_k x, K) \geq \sigma_{\min}(W_k).$$

677 Apply the least nonzero singular value of the unweighted boundary. The other diagonal
678 factor satisfies

$$679 \quad \sigma_{\min}(W_{k-1}^{-1}) \geq 1.$$

680 The Hodge decomposition applies these estimates separately to the down- and up-exact
681 sectors and gives

$$682 \quad \gamma_+(\Delta_{Y_a, d_a}(\lambda)) \geq c_a \lambda^{2d_a+4}.$$

683 A finite palette and $d_a = 2m_a - 1$, $m_a \leq m$, give the common bound Equation (6).

684 For padding, use augmented reduced chain complexes, including degree -1 , in
685 every join bidegree:

$$686 \quad \tilde{C}_d(Y_a * R_{\text{out}}) = \bigoplus_{r+s=d-1} \tilde{C}_r(Y_a) \otimes \tilde{C}_s(R_{\text{out}}).$$

687 On each summand the join Laplacian is the tensor sum

$$688 \quad \Delta_{Y_a * R_{\text{out}}} = \Delta_{Y_a} \otimes I + I \otimes \Delta_{R_{\text{out}}}.$$

689 The outside register has reduced homology only in its top degree and a constant
690 positive gap on every other mode. At the global target degree, therefore, the only
691 outside-harmonic bidegree is the active target degree tensored with $\mathcal{H}_{\text{out}}$. This proves
692 the stated zero-weight kernel and puts every other bidegree above a uniform floor.
693 Tensor the private output projector with $P_{\mathcal{H}_{\text{out}}}$; both outside differentials vanish after
694 this projection, preserving the private-pair bounds.

695 The same tensor-sum structure gives the padded form of Equation (6): the positive
696 eigenvalues of the padded Laplacian in degree d are sums $\mu_1 + \mu_2$ of factor eigenval-
697 ues. In every bidegree other than (d_a, s_{top}) the outside eigenvalue μ_2 is at least the
698 constant gap of R_{out} , and in the bidegree (d_a, s_{top}) a positive sum has $\mu_2 = 0$ and

699 $\mu_1 \geq \gamma_+(\Delta_{Y_a, d_a}(\lambda)) \geq c\lambda^\kappa$. Hence γ_+ of the padded degree- d Laplacian is at least
 700 $\min\{c\lambda^\kappa, \gamma_+(\Delta_{R_{\text{out}}})\} \geq c'\lambda^\kappa$ for $0 < \lambda \leq 1$.

701 Finally, the relative join formula identifies the padded connecting map with the
 702 active connecting map tensored with $\mathcal{H}_{\text{out}}$. Since all register weights are one, the
 703 positive-weight diagonal gauge fixes the register coordinates. Thus the padded bound-
 704 ary intersection is $\text{ran}(\pi_a \otimes I_{\mathcal{H}_{\text{out}}})$, completing the proof of Lemma 3.2.

705 A.2 Global decomposition and interface energy

706 If $A = \emptyset$, then $X_A = R$ and $K_A = V$, and Equation (8) holds with any $C \geq \gamma_R^{-1}$: in
 707 its top degree the register Laplacian $\Delta_R = \partial_R^* \partial_R$ is the tensor sum of the bowtie edge
 708 Laplacians, as in Appendix A.1, so it has kernel V and a constant positive floor γ_R on
 709 $V^\perp$, whence $\|(I - P_V)x\|^2 \leq \gamma_R^{-1} \langle x, \Delta_R x \rangle \leq \gamma_R^{-1} \langle x, \Delta_R x \rangle / (g_A \lambda^\kappa)$ because $g_A \leq 1$ and
 710 $\lambda \leq 1$. Assume henceforth $t_A \geq 1$.

711 Write the target chain space as the orthogonal coordinate sum

$$712 \quad C_d(X_A) = C_d(R) \oplus \bigoplus_{j \in A} C_{d,j}^{\text{priv}}.$$

713 For a unit chain x , write

$$714 \quad x = \omega_0 + \sum_j \omega_j, \quad x_j = \omega_0 + \omega_j, \quad U = \sum_j \|\omega_j\|^2, \quad e = \langle x, \Delta_A x \rangle.$$

715 Let $a = \partial_R \omega_0$, $y_j = L_j \omega_j$, $y = \sum_j y_j$, and $r = a + y$. The vector r is the shared-register
 716 component of the global down-boundary. Since $a + y_j = r - (y - y_j)$,

$$717 \quad \sum_j \|a + y_j\|^2 \leq 2t_A \|r\|^2 + 2 \sum_j \|y - y_j\|^2.$$

718 The elementary identity

$$719 \quad \sum_j \|y - y_j\|^2 = (t_A - 2) \|y\|^2 + \sum_j \|y_j\|^2$$

720 also holds for $t_A = 1$. Put

$$721 \quad \rho = \lambda^{-2} \|[L_1 \cdots L_{t_A}]\|^2 \leq N_* t_A, \quad \chi = (t_A - 2)_+ \rho + N_* \leq C t_A^2.$$

722 Then Equation (7) gives

$$723 \quad \sum_j \|y - y_j\|^2 \leq \chi \lambda^2 U.$$

724 All nonregister down-boundary coordinates and all up-boundary coordinates are private
 725 and partition by gadget. We obtain

$$726 \quad \sum_j \|D_j(\lambda) x_j\|^2 \leq 2t_A e + 2\chi \lambda^2 U. \quad (25)$$

727 A.3 Zero-weight and private-sector coercivity

728 Let Q_j also denote the orthogonal projector onto the padded zero-weight private
 729 subspace $Q_j \otimes \mathcal{H}_{\text{out}}$ of gadget j , and let $A_{j,0} = I - P_V - Q_j$ in the padded local chain
 730 space; by Lemma 3.2 this is the projector onto the orthogonal complement of $\ker D_{j,0}$.
 731 The positive singular-value floor of $D_{j,0}$ on $\text{ran } A_{j,0}$, together with $D_j(\lambda) = D_{j,0} + \lambda D_{j,1}$,
 732 gives

$$\begin{aligned}
 733 \quad S_A &:= \sum_j \|A_{j,0}x_j\|^2 \leq C \sum_j \|D_{j,0}x_j\|^2 \\
 734 &\leq C \left(t_A e + \chi \lambda^2 U + \lambda^2 \sum_j \|x_j\|^2 \right) \\
 735 &\leq C \left(t_A e + \chi \lambda^2 U + t_A \lambda^2 \right). \tag{26}
 \end{aligned}$$

737 The final inequality uses $\sum_j \|x_j\|^2 = t_A \|\omega_0\|^2 + U \leq t_A$.

738 Because $T_j P_V = 0$,

$$739 \quad T_j Q_j x_j = T_j x_j - T_j A_{j,0} x_j.$$

740 The lower bound on T_j restricted to Q_j , the estimate $\|T_j\| \leq C\lambda$, and orthogonality
 741 of the retained private output blocks imply

$$742 \quad S_Q := \sum_j \|Q_j x_j\|^2 \leq C \left(\frac{e}{\lambda^2} + S_A \right). \tag{27}$$

743 The orthogonality here concerns the retained T_j outputs. The shared L_j outputs were
 744 handled by their horizontal operator norm and need not be orthogonal.

745 Let $N = \|(I - P_V)x\|^2$. Summing the orthogonal decompositions $P_V \oplus Q_j \oplus \text{ran } A_{j,0}$
 746 gives

$$747 \quad N \leq U + t_A \|(I - P_V)\omega_0\|^2 = S_A + S_Q.$$

748 Since $U \leq N$, substituting Equations (26) and (27) and using $\chi \leq Ct_A^2$ yields

$$749 \quad N \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2})e \right] + Ct_A^2 \lambda^2 N.$$

750 Choosing $\lambda \leq c_0/t_{\max}$ absorbs the last term and proves Equation (11).

751 A.4 Logical coercivity

752 For each padded gadget Y_j , exact filling gives $\text{ran } \Pi_j \subseteq B_d(Y_j)$. The up-Laplacian
 753 $\partial_{d+1} \partial_{d+1}^*$ of Y_j has kernel $B_d(Y_j)^\perp$, and its positive eigenvalues are positive eigenvalues
 754 of the padded degree- d Laplacian, which the padded form of Equation (6) bounds
 755 below by $c\lambda^\kappa$. Therefore, after the common embedding into X_A ,

$$756 \quad \Delta_j^\uparrow \succeq c\lambda^\kappa \Pi_j.$$

757 The top-dimensional register has no up-columns, and every degree- $(d+1)$ simplex
 758 with a private vertex belongs to exactly one gadget. Hence the global up-Laplacian is
 759 the exact sum of the embedded local up-Laplacians, proving Equation (12).

Since $K_A \subseteq V$, the squared distance decomposes as

$$\|(I - P_{K_A})x\|^2 = N + \langle x, (P_V - P_{K_A})x \rangle.$$

Equations (5) and (12) give

$$\langle x, (P_V - P_{K_A})x \rangle \leq \frac{e}{cg_A\lambda^\kappa}.$$

Together with Equation (11),

$$\|(I - P_{K_A})x\|^2 \leq C \left[t_A\lambda^2 + (t_A + \lambda^{-2})e + \frac{e}{g_A\lambda^\kappa} \right].$$

For $\lambda \leq c_0/t_{\max}$, $g_A \leq 1$, and $\kappa \geq 2$, the last energy term dominates the middle energy terms up to a palette constant. This proves Equation (8).

A.5 Exact multiplicity and the whole-positive-spectrum gap

Let $\mathcal{L}_A(E_A)$ be the span of all eigenvectors of Δ_A with eigenvalue below E_A . Every unit $x \in \mathcal{L}_A(E_A)$ obeys $e < E_A$. Choosing the constants in Equation (9) makes the right-hand side of Equation (8) strictly less than $\eta^2 < 1$. Therefore $P_{K_A}x \neq 0$ for every nonzero $x \in \mathcal{L}_A(E_A)$, so $P_{K_A}|_{\mathcal{L}_A(E_A)}$ is injective and

$$\dim \mathcal{L}_A(E_A) \leq \dim K_A. \quad (28)$$

Every $v \in V$ remains a cycle after attaching the gadgets. By Lemma 4.1, the map $V/W_A \rightarrow H_d(X_A)$ is injective. Its domain has dimension $\dim K_A$, so Hodge theory supplies at least $\dim K_A$ zero eigenvectors of Δ_A . These zero vectors belong to $\mathcal{L}_A(E_A)$. Combining this lower bound with Equation (28) forces equality throughout: the geometric kernel has dimension $\dim K_A$, and no positive eigenvalue lies below E_A .

If $K_A = 0$, any unit vector in $\mathcal{L}_A(E_A)$ would have distance one from K_A , contradicting the strict $\eta < 1$ bound. Thus the same argument proves $\Delta_A \succeq E_AI$ in the zero-kernel case. No additional global relative-acyclicity assumption is required.

A.6 Proof of quotient naturality

Equation (14) makes q_A well-defined and injective. Its domain has dimension

$$\dim(V/W_A) = \dim K_A.$$

Hodge theory and Equation (10) give

$$\dim H_d(X_A) = \dim \ker \Delta_A = \dim K_A,$$

so q_A is also surjective. If $A \subseteq B$, both paths in the diagram of Theorem 4.2 send $v + W_A$ to the class of the same register cycle v in X_B . Thus the diagram commutes. The quotient map is onto because $W_A \subseteq W_B$, and its rank is $\dim(V/W_B) = \dim K_B$.

790 **B History-gap and source calculations**

791 **B.1 Rational gain/loss split**

792 On a clock bit c and work bit w , define

$$\begin{aligned} 793 \quad u_0 &= (|00\rangle + |01\rangle - |10\rangle)/\sqrt{3}, \quad u_1 = (|00\rangle - |01\rangle - |11\rangle)/\sqrt{3}, \\ d_0 &= (|00\rangle - |10\rangle - |11\rangle)/\sqrt{3}, \quad d_1 = (|01\rangle - |10\rangle + |11\rangle)/\sqrt{3}. \end{aligned}$$

794 The vectors in each pair are orthonormal. If consecutive history amplitudes are
 795 f_t, f_{t+1} , orthogonality to u_0, u_1 is equivalent to $f_{t+1} = \sqrt{2}Hf_t$; orthogonality to d_0, d_1
 796 is equivalent to $f_{t+1} = Hf_t/\sqrt{2}$. Applying the first relation to one work bit and the
 797 second to the other gives $H \otimes H$.

798 **B.2 Weighted path estimate**

799 With the unary encoding $|t\rangle = |1^t 0^{T-t}\rangle$, the legal clock span and its orthogonal
 800 complement are invariant under the neighboring-clock-guarded propagation, input,
 801 and output projectors. The diagonal clock term

$$802 \quad H_{\text{clock}} = \sum_{j=1}^{T-1} |01\rangle\langle 01|_{j,j+1}$$

803 has energy at least one on every illegal computational-basis clock string. By Hermiticity,
 804 the whole unary-clock Hilbert space is the orthogonal sum of the invariant legal and
 805 illegal sectors. It therefore suffices to prove the smaller inverse-polynomial bound
 806 on the legal sector. For a legal-clock state, write $f_t = s_t V_t \xi_t$, with the notation
 807 of Equation (18). Its norm and initial-history energy are

$$808 \quad N(\xi) = \sum_t w_t \|\xi_t\|^2, \quad E(\xi) = \sum_{t=1}^{L-1} c_t \|\xi_t - \xi_{t-1}\|^2 + \langle \xi_0, A_{\text{in}} \xi_0 \rangle,$$

809 where $w_t = s_t^2 \in [1, 2]$, $c_t \in \{1/2, 2/3\}$, and A_{in} is the clean-input penalty at time zero,
 810 a sum of commuting computational-basis projectors, one per clean qubit. It is not in
 811 general a projector: a basis string violating k clean bits has energy k . All that is used
 812 below is

$$813 \quad \ker A_{\text{in}} = C, \quad A_{\text{in}} \succeq P_{C^\perp}.$$

814 Split every ξ_t into the clean subspace C and its orthogonal complement, and split the
 815 energy accordingly as $E(\xi) = E_C + E_B$ with

$$816 \quad E_C = \sum_{t=1}^{L-1} c_t \|\xi_t^C - \xi_{t-1}^C\|^2, \quad E_B = \sum_{t=1}^{L-1} c_t \|\xi_t^{C^\perp} - \xi_{t-1}^{C^\perp}\|^2 + \langle \xi_0, A_{\text{in}} \xi_0 \rangle;$$

817 the two difference terms are orthogonal because C is invariant under the splitting.

818 For a state orthogonal to all valid histories, the clean component has weighted
 819 mean zero:

$$820 \quad \sum_t w_t \xi_t^C = 0.$$

821 The weighted variance identity and path Cauchy–Schwarz inequality give

$$822 \quad \sum_t w_t \|\xi_t^C\|^2 = \frac{1}{2 \sum_t w_t} \sum_{i,j} w_i w_j \|\xi_i^C - \xi_j^C\|^2 \leq 2L^2 E_C.$$

823 For the dirty component, set $a = \|\xi_0^{C^\perp}\|^2 \leq \langle \xi_0, A_{\text{in}} \xi_0 \rangle \leq E_B$ and $D_B = \sum_t \|\xi_t^{C^\perp} - \xi_{t-1}^{C^\perp}\|^2 \leq 2E_B$, using $c_t \geq 1/2$. The bound

$$825 \quad \|\xi_t^{C^\perp}\|^2 \leq 2a + 2(L-1)D_B$$

826 and $\sum_t w_t \leq 2L$ imply

$$827 \quad \sum_t w_t \|\xi_t^{C^\perp}\|^2 \leq 2L(2E_B + 4(L-1)E_B) \leq 8L^2 E_B.$$

828 Adding both sectors proves $N(\xi) \leq 8L^2 E(\xi)$ on the orthogonal complement of the
829 complete initial kernel, establishing the first bound in Equation (19).

830 B.3 Adding the output projector

831 We record the two-positive-operator estimate used for the second bound in Equation
832 (19). Let P project onto $\ker A$, suppose $A \succeq g(I - P)$, and let R be an orthogonal
833 projector. Put $\mathcal{S} = \ker A \cap \ker R$, and suppose $PRP \succeq \alpha P$ on $\ker A \ominus \mathcal{S}$. On every
834 nontrivial canonical block for the pair P, R , for some $\rho \in [\alpha, 1]$,

$$835 \quad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} \rho & \sqrt{\rho(1-\rho)} \\ \sqrt{\rho(1-\rho)} & 1-\rho \end{pmatrix}.$$

836 Thus $g(I - P) + R$ has trace $g + 1$, determinant $g\rho$, and smallest eigenvalue

$$837 \quad \frac{g + 1 - \sqrt{(g + 1)^2 - 4g\rho}}{2}.$$

838 This quantity increases with ρ ; the one-dimensional canonical blocks outside $\mathcal{S}$ obey
839 the same lower bound. Since $A + R \succeq g(I - P) + R$ and both operators have kernel
840 $\mathcal{S}$,

$$841 \quad \gamma_+(A + R) \geq \frac{g + 1 - \sqrt{(g + 1)^2 - 4g\alpha}}{2} \geq \frac{g\alpha}{g + 1}. \quad (29)$$

842 For the history Hamiltonian, $g = 1/(8L^2)$ and $\alpha = (1 - r)/Z \geq 1/(3L)$ when $r \leq 1/3$.
843 Substitution into Equation (29) gives

$$844 \quad \gamma_+(H_{\text{out}}) \geq \frac{1}{3L(8L^2 + 1)}.$$

845 The kernel identity follows directly from positivity:

$$846 \quad \ker(H_{\text{in}} + H_{\text{rej}}) = \ker H_{\text{in}} \cap \ker H_{\text{rej}} = \mathcal{V} \ker(I - M).$$

847 B.4 Exact eight-label compression

848 Let U_x denote the fixed-input verifier and q its output bit. The label z is never changed.
 849 After computing and uncomputing the constant-size acceptance predicate, compression
 850 through all clean work registers has no matrix element between distinct labels. On
 851 $z = 000$, acceptance is the identity. On five labels it is the scalar $p_x = \|P_{\text{acc}}U_x|0 \cdots 0\rangle\|^2$,
 852 and on two labels it is zero. This proves Equation (22) even for a coherent vector
 853 across label values.

854 The normalized trace is

$$855 \quad \frac{\text{Tr } M_x}{8} = \frac{1 + 5p_x}{8}.$$

856 If $p_x = 1$, this equals $3/4$ and coincides with the perfect-space fraction. If $p_x \leq 1/3$, it
 857 is at most $1/3$, the perfect-space fraction is exactly $1/8$, and every off-perfect eigenvalue
 858 is at most $1/3$. Thus the construction simultaneously supplies the trace promise, the
 859 perfect-space separation, and the off-perfect ceiling needed by the output-gap proof.

860 B.5 Source-interface boundary

861 The exact source used by Theorem 5.3 begins with a fixed-input $\text{BQP}_1^{G_2}$ verifier. The
 862 three mixed label bits are added by the reduction and are the entire input space of
 863 Equation (22); the verifier's input and all predicate scratch remain clean. Consequently
 864 the initial history dimension is exactly eight. No trace-to-rank inference, unrestricted
 865 SDQC_1 identification, or approximate gate compilation enters the proof.

866 The finite gadget evidence needed to compile these circuit terms is given in Ap-
 867 pendix C. The theorem uses those exact finite objects and does not infer their topology
 868 from floating-point spectra.

869 For an arbitrary acceptance operator with $M|_{S^\perp} \preceq rI$, put $f = \dim S/D$ and
 870 $p = \text{Tr}(M)/D$. Spectral decomposition gives

$$871 \quad f \leq p \leq r + (1 - r)f.$$

872 Thus trace promises $p \geq a$ versus $p \leq b$ force a perfect-space fraction gap only when
 873 $\max\{0, (a - r)/(1 - r)\} > b$. The eight-label source avoids this obstruction by fixing
 874 the perfect fractions directly.

875 An exact circuit over $\{X, CX, CCX, H\}$ admits the following real-gate extension.
 876 Add one maximally mixed, unmeasured spectator a and replace each H_j by $H_j \otimes H_a$.
 877 If there are h Hadamards, then

$$878 \quad U' = U \otimes H_a^h, \quad J' = J \otimes I_a, \quad P'_{\text{acc}} = P_{\text{acc}} \otimes I_a.$$

879 For either parity of h ,

$$880 \quad (J')^*(U')^*P'_{\text{acc}}U'J' = M \otimes I_a.$$

881 Both the input dimension and perfect eigenspace dimension double, while the fraction
 882 and off-perfect ceiling are unchanged. This calculation does not cover complex phases.

C Finite palette and guard certificates

This appendix identifies the finite objects used in Lemma 5.1. The underlying active graphs are four explicit constructors in the source artifact: `state_0m1`, `state_00m11`, `state_01m10`, and `state_00m10m11`. They accompany Rudolph’s gain/loss construction [22]. The supplementary certificate archive records the complete graphs, oriented integer fillings, kernel bases, modular rank witnesses, and Gram determinants. No numerical eigenvalue threshold is used.

Active vector	degree	vertices	dim V	dim Q	certificate mod p
$ 0\rangle - 1\rangle$	1	16	2	8	$\det G = 512$
$ 00\rangle - 11\rangle$	3	33	4	112	$\det G = 655970$
$ 01\rangle - 10\rangle$	3	33	4	112	signed relabeling
$ 00\rangle - 10\rangle - 11\rangle$	3	41	4	176	$\det G = 797443$
three remaining gain/loss vectors	3	41	4	176	signed relabelings

Table 1 Exact active-palette data. Vertex counts include the $7m_a$ register vertices of the active bowties. In each determinant row, $p = 1000003$ and $G = T_{a,1}^* T_{a,1}$ on Q_a ; a nonzero residue proves injectivity over $\mathbb{Q}$.

For the one-qubit atom, exact boundary ranks are 15, 24 and the Betti numbers are $(1, 1, 0)$; a denominator-one 24-term two-chain fills the target cycle. At zero weight the target differential pair has rank 30, and the two register cycles plus eight private vectors give its complete ten-dimensional kernel. For $|00\rangle - |11\rangle$, the clique counts in degrees zero through four are $(33, 241, 646, 712, 272)$, the boundary ranks are $(32, 209, 437, 272)$, and the Betti numbers are $(1, 0, 0, 3, 0)$. A denominator-one 248-term four-chain fills the target; rank 596 of the zero-weight pair matches the four register and 112 private kernel vectors. Swapping the second bowtie petals transports this certificate exactly to $|01\rangle - |10\rangle$.

For the three-term representative, the clique counts in degrees zero through five are

$$(41, 322, 914, 1082, 477, 30),$$

and the boundary ranks are $(40, 282, 632, 447, 30)$. Hence the target degree-three Betti number is three. A denominator-one 382-term four-chain fills $|00\rangle - |10\rangle - |11\rangle$, while three complementary register cycles remain independent modulo boundaries. The zero-weight pair has rank 902, matching its kernel $V \oplus Q$ of dimension $4 + 176$. Signed permutations of bowtie petals transport the graph, filling, kernel, and private pair to the other three gain/loss vectors.

C.1 Selected-cycle guards

We record the guard closure because it is part of the theorem interface, not an appeal to a general integer-state gadget. Let Y implement π_ϕ on register R , let B be a fresh

bowtie with petals S_0, S_1 , and choose $\beta \in \{0, 1\}$. Add all edges of B , join every vertex of R to every vertex of B , and join each private vertex of Y only to the selected petal S_β . The resulting flag complex is

$$Z = (Y * S_\beta) \cup (R * B), \quad (Y * S_\beta) \cap (R * B) = R * S_\beta. \quad (30)$$

Its new register is $R' = R * B$. If γ_β is the normalized oriented cycle of the selected petal, then Z implements

$$\pi_\phi \otimes |\beta\rangle\langle\beta|, \quad Q' = Q \otimes \mathbb{C}\gamma_\beta.$$

Indeed, reduced relative chains give

$$\tilde{C}_*(Z, R') \cong \tilde{C}_*(Y, R) \hat{\otimes} \tilde{C}_*(S_\beta).$$

The connecting map therefore tensors the old exact boundary intersection with $[\gamma_\beta]$, proving exact filling and no additional killed register class. At zero private weight the private differential is the corresponding tensor-sum differential. Since S_β has reduced homology only in degree one, the complete target kernel is $V' \oplus Q'$; every other bidegree pays a fixed positive guard gap. Finally, project the old private output pair onto the selected harmonic line. Both outside differentials vanish under this projection, so

$$T_Z(\lambda) = \lambda T_{Z,1}, \quad T_{Z,1}V' = 0, \quad \|T_{Z,1}(q \otimes \gamma_\beta)\| \geq b\|q \otimes \gamma_\beta\|.$$

Removing the unique old private vertex also gives $LL^* \preceq N_{\text{priv}}\lambda^2 I$. Iterating this construction for the constant number of required basis guards preserves a fixed finite palette.

The basis projector itself is the special case $R \cup (v * S_\beta)$, where a private apex v cones the selected petal. Its private zero-weight kernel Q is zero, so the private-pair condition is vacuous. Together with Table 1, this proves Lemma 5.1.

C.2 Reproducibility boundary

The supplementary archive contains the exact witnesses in

- `RUDOLPH_REPRESENTATIVE_BULK_CERTIFICATE.json`,
- `REMAINING_ACTIVE_ATOM_CERTIFICATES.json`,
- `ACTIVE_HADAMARD_ORBIT_CERTIFICATE.json`, and
- `SELECTED_CYCLE_GUARD_CHECKS.json`.

Their checkers recompute the stated ranks, fillings, relabelings, and guard identities. These finite checks certify the displayed palette; the analytic implication from those certificates is the proof of Theorem 3.3.

AI Disclosure

We used ChatGPT and Claude as auxiliary tools for literature search, brainstorming, proof checking, and drafting and editing of the exposition (Sections 1, 3–8, and Appendices A–C). The authors evaluated and revised all AI-assisted material and take full responsibility for the correctness and originality of the submission, including every proof and reference.